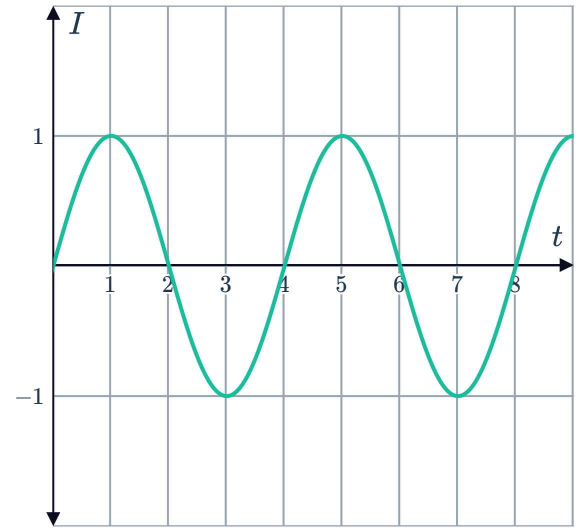
Applications

1 Sounds around us create pressure waves. Our ears interpret the amplitude and frequency of these waves to make sense of the sounds.
A speaker is set to create a single tone, and the graph below shows how the pressure intensity (I) of the tone, relative to atmospheric pressure, changes over t seconds:

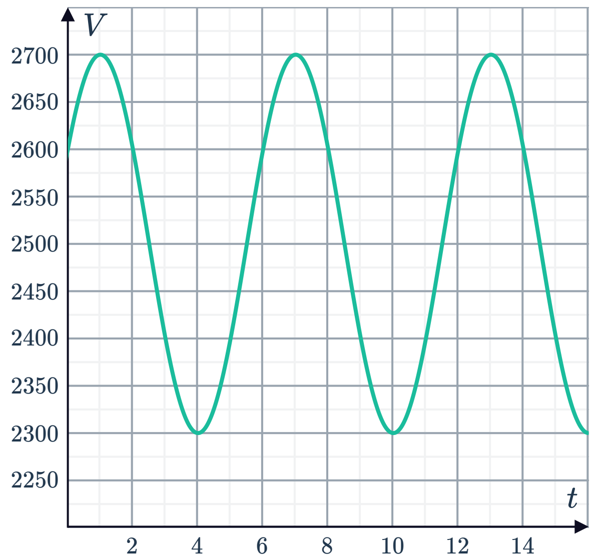


a State whether a sine or cosine function be more suited for modelling this graph. Explain your answer.

b Find the equation of the function for the intensity I after t seconds.

c As a brain exercise, a music student must tap their foot every time they hear the most intense part of the sound. How many times will the student tap their foot in 14 minutes?

2 Marge has asthma, so she is having some breathing tests done to monitor her lung capacity V measured in millilitres (mL). The respirologist models Marge's breathing patterns according to the model $V = a\sin\left(b\left(t+c\right)\right)+d$, where the quantity $b\left(t+c\right)$ is in degrees. The graph of the function is shown below:



a How long is one full breath (inhale and exhale) in seconds?

b Find the value of d, the median amount of air in Marge's lungs.

c Find the value of a, the difference between the maximum and median amount of air.

d Find the value of b.

e If the initial volume of air in Marge's lungs was 2600 mL, find the value of c.

3 A circular Ferris wheel that is 40 m in diameter, contains several carriages. Hannah and Michael enter a carriage at the bottom of the Ferris wheel, and get off 6 minutes later after having gone around completely 3 times. When a carriage is at the bottom of the wheel, it is 1 m above the ground.

a State whether the functions below could be used to model Hannah and Michael's height, h, above the ground t minutes after getting on:

i $h(t) = A(t-a)^2 + k$

ii $h(t) = Ae^{kt} + B$

iii $h(t) = A\sin(kt) + B$

iv $h(t) = A\cos(kt) + B$

b For the height function $h(t)$, find:

i The period.

ii The minimum height above ground.

iii The maximum height above ground.

c Sketch the graph of $h(t)$ for the 6 minute ride.

d State the equation for the height function $h(t)$.

e Find the first time, t minutes, at which the passenger reaches a height of 21 metres above the ground.

f A new function is formed by translating $h(t)$ down by 21 m. Describe what this new function represents.

4 The height above the ground of a rider on a Ferris wheel can be modelled by the function:

$$h(t) = 25\cos(3(t-60)) + 30$$

where $h(t)$ is the height in metres, t is the time in seconds, and the quantity $(3(t-60))$ is in degrees.

a Sketch a graph of $h(t)$ for $0^\circ \leq t \leq 240^\circ$.

b Find the maximum height of the rider.

c Find the minimum height of the rider.

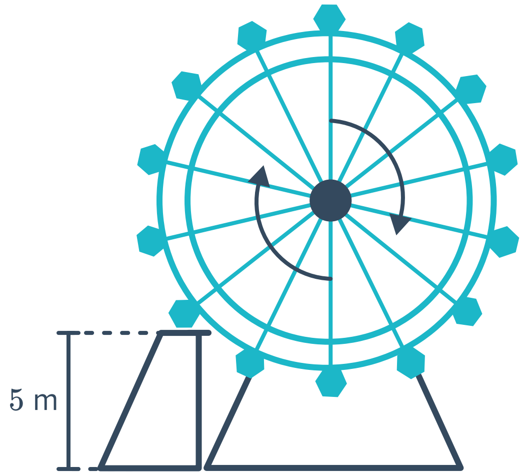
d Find the height of the rider at $t = 85$ seconds. Round your answer to two decimal places.

e Find the time taken for the rider to complete one full revolution.

5 Passengers on a Ferris wheel access their seats from a platform 5 m above the ground. As each seat is filled, the Ferris wheel moves around clockwise so that the next seat can be filled. Once all seats are filled, the ride begins and lasts for 6 minutes. The height h in metres of Amelia's seat above the ground t seconds after the ride has begun is given by:

$$h(t) = 14\sin(10t - 40) + 16$$

where the quantity $(10t - 40)$ is in degrees.



a Sketch a graph of the function $h(t)$ for one rotation.

b Find the height above the ground of Amelia's seat at the commencement of the ride. Round your answer to two decimal places.

c Find the time at which Amelia first passes the access platform after the ride commences. Round your answer to two decimal places.

d Find the number of times her seat passes the access platform in the first two minutes.

e Due to a malfunction, the Ferris wheel stops abruptly 1 minute and 40 seconds into the ride. Find the height above the ground that Amelia is stranded at. Round your answer to two decimal places.

6 The height in metres of the tide above mean sea level t hours after midnight is given by:

$$h = 4\sin(30(t - 2))$$

a Find the height above mean sea level of the tide at 1:00 am.

b Find the time of the first high tide of the day.

c Find the number of hours between a high tide and the following low tide.

d Sketch a graph of h over the first 24 hours.

e Find the height the tide is predicted to be at 2:00 pm.

f How much higher than low tide level is the tide at 11:30 am? Round your answer to two decimal places.

7 The tide rises and falls in a periodic manner which can be modelled by a trigonometric function. Brad charts tide levels in order to determine when he can sail his ship into a bay and takes the following measurements:

- Low tide occured at 8 am, when the bay was 7 m deep.
- High tide occured at 2 pm, when the bay was 15 m deep.

a Let t be the number of hours passed since low tide was first measured. Complete the following table of values for the tide level:

Time $(t, \text{ hours})$	0	3	6	9	12
Tide level $(d, \text{ metres})$					

b Find the amplitude of the tide level.

c Find the period of the tide level.

d Sketch a graph of the tide level, d, over the first 12 hours.

e State whether the following equations can be used to model the tide level, d, at t hours after 8 am, where the parameters A, n and C are positive:

i $d = -A\cos nt + C$

ii $d = A\cos nt + C$

iii $d = A\sin nt + C$

iv $d = -A\sin nt + C$

f Find an equation for d.

g The ship needs a depth of at least 11.7 metres to be able to sail into the bay safely. Find the number of hours Brad has after high tide before he can no longer safely sail into the bay. Round your answer to two decimal places.

8 The height of water in metres on a beach wall t hours after midnight is given by:

$$h(t) = 6 + 4\cos(30t - 60)$$

a Write the equation in the form $h(t) = 6 + 4\cos(30(t - c))$, where c is the least positive number.

b Sketch a graph of $h(t)$ over the first 24 hours.

c Find the earliest time of day at which the water is at its highest.

d Find the time when the water level is first 2 m up the wall.

9 In a tidal river, the time between high tide [redacted] hours. The average depth of water at a point in the river is 5 m, and at high tide the depth is 8 m. The depth of water at this point, t hours after noon, is given by:

$$h(t) = a\sin(bt + c) + d$$

where $a > 0$, c is the least positive number, the quantity $(bt + c)$ is in degrees.

a Given that high tide occurs at noon, find the value of:

i d ii a iii b iv c

b Sketch a graph of $h(t)$ over the first 24 hours.

c Within the first 24 hours, at what times is the depth of the water 2 m?

10 Tobias is jumping on a trampoline. Victoria watches him bounce at a regular rate and wants to try to model his height over time. When Victoria starts her stopwatch, Tobias is at a minimum height of 30 cm below the trampoline frame. A moment later Victoria records Tobias reaching a maximum height of 50 cm above the trampoline frame. To model the situation Victoria uses the function:

$$H(t) = a\sin(360(t - c)) + d$$

where H is the height in centimetres above the trampoline frame and t is the time in seconds and $0 < c < 1$.

a Find the value of:

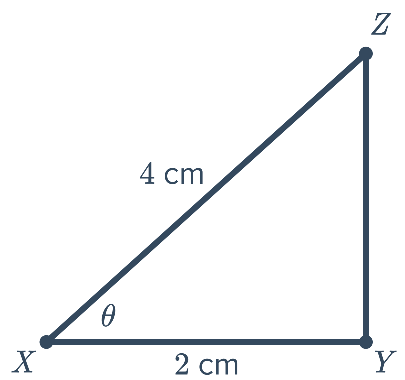
i a ii d iii c

b At what time does Tobias first reach a height of 30 cm above the trampoline frame?

11 Three objects, X, Y and Z are placed in a magnetic field such that:

- X is 2 cm from object Y.
- X is 4 cm from object Z.
- As object X is moved closer to line YZ, objects Y and Z move in such a way that the lengths XY and XZ remain fixed.

Let $\theta = \angle ZXY$, and let the area of triangle XYZ be represented by A.



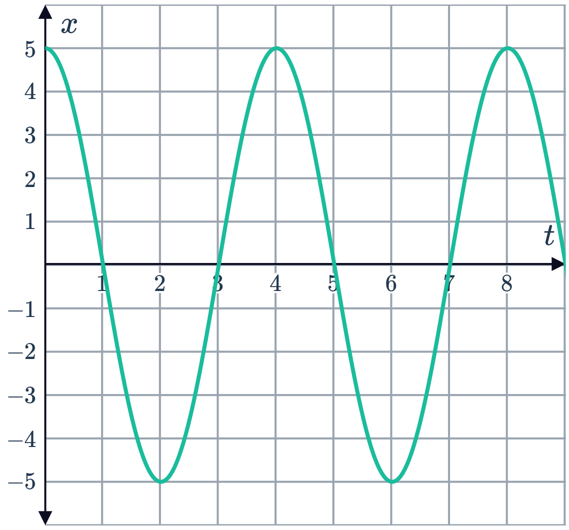
a Find an equation for A in terms of θ.

b State the domain of the function of A in this context.

c Sketch a graph of the function A.

d Find the acute angle which will form an area that is exactly half of the maximum possible area.

12 A metronome is a device used to help kee[illegible] beat consistent when playing a musical instrument. It swings back and forth between its end points, just like a pendulum.
For a particular speed, the given graph represents the metronome's distance, x cm, from the centre of its swing, t seconds after it starts swinging. Negative values of x represent swinging to the left, and positive values of x represent swinging to the right of the centre:



a Find the position of the pendulum when it first started swinging.

b From its starting position, did the pendulum start to swing to the left or to the right?

c Find the time at which the pendulum first passes the centre of its swing.

d Find the time taken to complete one full swing.

e Find the function for x in terms of t.